

v0.1 scalar vs v2 evidence-vector feedback in the agentic harness

Question: the v2 metric (pdf_fidelity research system) deliberately emits NO single score — a gate-first evidence vector. The harness needs feedback to hill-climb. Does feeding the agent the v2 vector (gates + independent drivers) produce better conversions than the v0.1 blended scalar it currently optimizes?

Design: same model (opus), effort (medium), visual on, \$2 cap, same samples. Two arms differ ONLY in the mid-loop feedback the agent sees:

ARM A v0.1 -> one blended overall = $\text{content}^{.35} * \text{visual}^{.65}$ (compensatory: one axis hides another)

ARM B v2 -> HARD GATES (page-count, token P/R, number-F1) then drivers (ink_f1, ltsim, center)

Grading is identical for both: the v2 evidence vector at 96 DPI (neither arm can game its own feedback).

How to read this — columns & method

HARD GATES (non-compensatory; a human reviewer objects if any fails). PASS/FAIL, not blended:

G1 page-count delta = 0 G2 token recall $\geq .95$ G3 token precision $\geq .95$ G4 number-F1 = 1 (when numbers present)

CONTINUOUS DRIVERS (hill-climb once gated; Phase A found these the most informative, mutually complementary):

ink_f1_reg registered tolerant-ink overlap (raster) ltsim_macro text-block transport center_q90 word displacement (lower=better)

driver_mean = mean(ink_f1, ltsim, 1-center_q90) — a convenience summary, NOT part of the v2 metric.

REPORT-ONLY (diagnostic, not optimized): strict_f1 (drops on math the PDF tokenizes oddly — not real loss),

number_f1, ssim_reg (abstains unless canvas matches), page_break_f1 (noisy — use page-count delta instead).

Baseline = 1-turn (no loop). Arm A / Arm B = agentic opus, medium, visual, harness — identical except feedback.

Scoreboard (all arms scored by v2)

bold = better of Arm A vs B per column, per sample; AVERAGE = mean over compiled runs

sample	arm	gates	ink_f1	ltsim	cq90	drv	numF1	pagD	cost	turns
06_tables_moderate_010	baseline	2/4	0.10	0.66	0.30	0.49	0.993	1	—	—
	A v0.1	3/4	0.62	0.78	0.08	0.77	0.995	0	\$1.65	28
	B v2	4/4	0.56	0.78	0.04	0.77	1.000	0	\$2.01	36
05_tables_simple_023	baseline	1/4	0.09	0.67	0.34	0.47	0.951	-1	—	—
	A v0.1	3/4	0.59	0.92	0.03	0.83	0.998	0	\$2.02	37
	B v2	4/4	0.46	0.87	0.09	0.75	1.000	0	\$2.03	40
09_algorithms_003	baseline	1/4	0.35	0.74	0.31	0.59	0.740	0	—	—
	A v0.1	1/4	0.56	0.85	0.03	0.79	0.989	0	\$1.96	34
	B v2	4/4	0.49	0.81	0.06	0.75	1.000	0	\$2.03	34
pubmed_table_004	baseline	1/4	0.02	0.64	0.30	0.45	0.719	-1	—	—
	A v0.1	3/4	0.59	0.90	0.02	0.82	0.997	0	\$1.57	28
	B v2	4/4	0.67	0.89	0.01	0.85	1.000	0	\$1.75	32
arxiv5t_paper_019	baseline	nc	—	—	—	—	—	—	—	—
	A v0.1	1/4	0.37	0.76	0.18	0.65	0.992	0	\$1.88	28
	B v2	1/4	0.34	0.77	0.21	0.63	0.992	0	\$2.01	31
i2s_equation_001	baseline	nc	—	—	—	—	—	—	—	—
	A v0.1	1/4	0.44	0.74	0.08	0.70	0.923	0	\$1.26	30
	B v2	2/4	0.32	0.74	0.05	0.67	1.000	0	\$1.96	32
AVERAGE	baseline	1.2/4	0.14	0.68	0.31	0.50	0.851			
	A v0.1	2.0/4	0.53	0.82	0.07	0.76	0.982			
	B v2	3.2/4	0.47	0.81	0.08	0.74	0.999			

Case: 06_tables_moderate_010 — Table reflow, 2->3 pages (lathe hard)

gates A 3/4 vs B 4/4

REFERENCE (pdfLaTeX)ARM A: v0.1 scalar feedbackARM B: v2 evidence-vector feedback

Moderate Tables

Source-backed grouped table sample

1 Tables

These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

Table 1: Source table 1: 2512.03369_table_1

Dataset	Coverage	Task	Spn. Hms.	Temp. Hms.	Area (km²)	Period	Fig./Video	Device	Mod.
SizeRoad-Fire	Worldwide	Fire Mask Forecast	30 m	1 hour	20,000,000	2013-2021	Fig	Satellite	5
FireSpentITS	USA	Fire Mask Backtrack	375 m	1 day	9,834,000	2018-2021	Fig	Satellite	4
Mongoon	Mediterranean	Fire Mask Forecast	1000 m	1 day	9,000,999	2006-2022	Fig	Satellite	4
FireDBI	USA	Fire Intensity Forecast	375 m	1 day	9,834,000	2012-2017	Fig	Satellite	4
Next Day fire	USA	Fire Mask Forecast	1 km	1 day	9,834,000	2012-2020	Fig	Satellite	4
SmallFire Colls	Worldwide	Fire Behavior	500 m	1 day	149,000,000	2003-2017	Fig	Satellite	4
CPDS	Canada	Fire Behavior	180 m	1 day	9,985,000	2002-2021	Fig	Satellite	4
BA-CONFIRE	Worldwide	Fire Behavior	111 km	1 month	149,000,000	1950-2021	Fig	Satellite	1
GlobalFire	Worldwide	Fire Behavior	500 m	1 day	149,000,000	2001-2017	Fig	Satellite	1
Re-FWI	Canada	Fire Behavior	80 km	1 day	9,985,000	1980-2018	Fig	Satellite	7
GFIS	Worldwide	Fire Behavior	30 m	16 days	149,000,000	2003-2016	Fig	Satellite	1
FFA FFD-A	USA	Fire Behavior	4 km	1 day	9,834,000	1990-2020	Fig	Satellite	4
GFED5	Worldwide	Fire Behavior	500 m	1 month	149,000,000	1997-2020	Fig	Satellite	1
CACAB	Worldwide	Fire Behavior	27.7 km	1 month	149,000,000	2002-2020	Fig	Satellite	3
PT-FirePop	Portugal	Fire Mask Forecast	4 km	14 hours	92,150	2015-2021	Fig	Satellite	1
NSMC-IR	China	Fire Monitoring	3 km	1 hour	9,597,000	2019-2021	Fig	Satellite	1
WCV-US	Western USA	Danger Forecast	250 m	5 weeks	4,200,000	2005-2017	Fig	Satellite	1
ECUF	Region in South Korea	Danger Forecast	1 km	1 month	605	2017-2021	Fig	Satellite	5
LCQJ	Region in China	Causality Prediction	500 m	12 hour	16,411	2015-2020	Fig	Satellite	4
Calchar	Region in USA	Fire Mask Forecast	20 m	1 year	450,000	2015-2022	Fig	Satellite	1
GABAM	Worldwide	Fire Behavior	30 m	1 year	149,000,000	1990-2021	Fig	Satellite	1
Fire Atlas	Worldwide	Fire Behavior	500 m	1 day	149,000,000	2003-2016	Fig	Satellite	1
MODIS	Worldwide	Fire Mask Forecast	1 km	1 day	149,000,000	2000-2021	Fig	Satellite	3
VIIRS	Worldwide	Fire Mask Forecast	375 m	12 hours	149,000,000	2012-2024	Fig	Satellite	3
NOAA IRIS Fire	North America	Danger Forecast	2 km	1 day	24,710,000	2003-2024	Fig	Satellite	3
NOAA IRIS Smoke	North America	Danger Forecast	2 km	1 day	24,710,000	2003-2024	Fig	Satellite	3
GORE live	Western Hemisphere	Fire Mask Forecast	2 km	5 mins	61,000,000	2017-2024	Fig	Satellite	4
NIW WF	USA	Fire Mask Forecast	2 km	5 mins	9,834,000	2000-2024	Fig	Satellite	1
FHY	Worldwide	Fire Behavior	500 m	1 day	149,000,000	2005-2011	Fig	Satellite	1
FireCanada	Canada	Fire Behavior	1 km	1 day	61	2014.8	Fig	Satellite	3
FireSatrapy	Region in China	Fire Video Prediction	0.5 m	1 s	0.7	2023.2	Video	Drone	4

raster diff

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boxes drawn from v2 evidence coords: red = biggest raster/appearance diff (ink_f1, ssim) amber = most-displaced matched word (center_q90)

ARM A v0.1 gates 3/4

G1=P G2=P G3=P G4=F ink=0.62 ltsim=0.78 cq90=0.08 numF1=0.995

→ numbers wrong: missing 1,2

ARM B v2 gates 4/4

G1=P G2=P G3=P G4=P ink=0.56 ltsim=0.78 cq90=0.04 numF1=1.000

→ all gates pass; residual is visual polish (amber/red boxes)

REFERENCE (pdfLaTeX)

ARM A: v0.1 scalar feedback

ARM B: v2 evidence-vector feedback

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raster diff

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boxes drawn from v2 evidence coords: red red = biggest raster/appearance diff (ink_f1, ssim) amber amber = most-displaced matched word (center_q90)

ARM A v0.1 gates 3/4

G1=P G2=P G3=P G4=F ink=0.59 ltsim=0.92 cq90=0.03 numF1=0.998

→ numbers wrong: missing 1

ARM B v2 gates 4/4

G1=P G2=P G3=P G4=P ink=0.46 ltsim=0.87 cq90=0.09 numF1=1.000

→ all gates pass; residual is visual polish (amber/red boxes)

Case: 09_algorithms_003 — Algorithm block, math extraction (lathe hard)

gates A 1/4 vs B 4/4

REFERENCE (pdfLaTeX)

Algorithm 2 Initialization of the control and state variables

```
for  $t \in \{-1, \dots, -T_{\text{traceback}}\}$  do ▷ Retrieve the values of  $P_{u,t}$ 
   $P_{u,t} \leftarrow \text{Power.GetValue}(t)$ 
end for
for  $t \in \{-1, \dots, -T_{\text{traceback}}\}$  do ▷ Initial conditions on the state variables
  if  $P_{u,t} > \text{MinimumPower.GetValue}(t)$  then
     $S_{\text{OFF}} \leftarrow 0$ 
     $S_{\text{STOP}} \leftarrow 0$ 
     $S_{\text{START}} \leftarrow 0$ 
    if  $P_{u,t} < P_{u,t-1}$  then ▷ Exact initialization required only if the FLAT state is defined.
       $S_{\text{ON\_UP}} \leftarrow 0$ 
       $S_{\text{ON\_DOWN}} \leftarrow 1$ 
       $S_{\text{ON\_FLAT}} \leftarrow 0$ 
    else if  $P_{u,t} > P_{u,t-1}$  then
       $S_{\text{ON\_UP}} \leftarrow 1$ 
       $S_{\text{ON\_DOWN}} \leftarrow 0$ 
       $S_{\text{ON\_FLAT}} \leftarrow 0$ 
    else if  $P_{u,t} = P_{u,t-1}$  then
       $S_{\text{ON\_UP}} \leftarrow 0$ 
       $S_{\text{ON\_DOWN}} \leftarrow 0$ 
       $S_{\text{ON\_FLAT}} \leftarrow 1$ 
    end if
  else if  $P_{u,t} > 0$  then ▷ Reconstruct the startups and shutdowns
    if  $P_{u,t} < P_{u,t-1}$  then
       $S_{\text{STOP}} \leftarrow 1$ 
       $S_{\text{START}} \leftarrow 0$ 
    else
       $S_{\text{STOP}} \leftarrow 0$ 
       $S_{\text{START}} \leftarrow 1$ 
    end if
  else ▷ Final possibility: the unit is OFF.
     $S_{\text{OFF}} \leftarrow 1$ 
     $S_{\text{STOP}} \leftarrow 0$ 
     $S_{\text{START}} \leftarrow 0$ 
     $S_{\text{ON\_UP}} \leftarrow 0$ 
     $S_{\text{ON\_DOWN}} \leftarrow 0$ 
     $S_{\text{ON\_FLAT}} \leftarrow 0$ 
  end if
end for
```

ARM A: v0.1 scalar feedback

raster diff

Algorithm 2 Initialization of the control and state variables

```
for  $t \in \{-1, \dots, -T_{\text{traceback}}\}$  do ▷ Retrieve the values of  $P_{u,t}$ 
   $P_{u,t} \leftarrow \text{Power.GetValue}(t)$ 
end for
for  $t \in \{-1, \dots, -T_{\text{traceback}}\}$  do ▷ Initial conditions on the state variables
  if  $P_{u,t} > \text{MinimumPower.GetValue}(t)$  then
     $S_{\text{OFF}} \leftarrow 0$ 
     $S_{\text{STOP}} \leftarrow 0$ 
     $S_{\text{START}} \leftarrow 0$ 
    if  $P_{u,t} < P_{u,t-1}$  then ▷ Exact initialization required only if the FLAT state is defined.
       $S_{\text{ON\_UP}} \leftarrow 0$ 
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       $S_{\text{ON\_FLAT}} \leftarrow 0$ 
    else if  $P_{u,t} > P_{u,t-1}$  then
       $S_{\text{ON\_UP}} \leftarrow 1$ 
       $S_{\text{ON\_DOWN}} \leftarrow 0$ 
       $S_{\text{ON\_FLAT}} \leftarrow 0$ 
    else if  $P_{u,t} = P_{u,t-1}$  then
       $S_{\text{ON\_UP}} \leftarrow 0$ 
       $S_{\text{ON\_DOWN}} \leftarrow 0$ 
       $S_{\text{ON\_FLAT}} \leftarrow 1$ 
    end if
  else if  $P_{u,t} > 0$  then ▷ Reconstruct the startups and shutdowns
    if  $P_{u,t} < P_{u,t-1}$  then
       $S_{\text{STOP}} \leftarrow 1$ 
       $S_{\text{START}} \leftarrow 0$ 
    else
       $S_{\text{STOP}} \leftarrow 0$ 
       $S_{\text{START}} \leftarrow 1$ 
    end if
  else ▷ Final possibility: the unit is OFF.
     $S_{\text{OFF}} \leftarrow 1$ 
     $S_{\text{STOP}} \leftarrow 0$ 
     $S_{\text{START}} \leftarrow 0$ 
     $S_{\text{ON\_UP}} \leftarrow 0$ 
     $S_{\text{ON\_DOWN}} \leftarrow 0$ 
     $S_{\text{ON\_FLAT}} \leftarrow 0$ 
  end if
end for
```

ARM B: v2 evidence-vector feedback

raster diff

Algorithm 2 Initialization of the control and state variables

```
for  $t \in \{-1, \dots, -T_{\text{traceback}}\}$  do ▷ Retrieve the values of  $P_{u,t}$ 
   $P_{u,t} \leftarrow \text{Power.GetValue}(t)$ 
end for
for  $t \in \{-1, \dots, -T_{\text{traceback}}\}$  do ▷ Initial conditions on the state variables
  if  $P_{u,t} > \text{MinimumPower.GetValue}(t)$  then
     $S_{\text{OFF}} \leftarrow 0$ 
     $S_{\text{STOP}} \leftarrow 0$ 
     $S_{\text{START}} \leftarrow 0$ 
    if  $P_{u,t} < P_{u,t-1}$  then ▷ Exact initialization required only if the FLAT state is defined.
       $S_{\text{ON\_UP}} \leftarrow 0$ 
       $S_{\text{ON\_DOWN}} \leftarrow 1$ 
       $S_{\text{ON\_FLAT}} \leftarrow 0$ 
    else if  $P_{u,t} > P_{u,t-1}$  then
       $S_{\text{ON\_UP}} \leftarrow 1$ 
       $S_{\text{ON\_DOWN}} \leftarrow 0$ 
       $S_{\text{ON\_FLAT}} \leftarrow 0$ 
    else if  $P_{u,t} = P_{u,t-1}$  then
       $S_{\text{ON\_UP}} \leftarrow 0$ 
       $S_{\text{ON\_DOWN}} \leftarrow 0$ 
       $S_{\text{ON\_FLAT}} \leftarrow 1$ 
    end if
  else if  $P_{u,t} > 0$  then ▷ Reconstruct the startups and shutdowns
    if  $P_{u,t} < P_{u,t-1}$  then
       $S_{\text{STOP}} \leftarrow 1$ 
       $S_{\text{START}} \leftarrow 0$ 
    else
       $S_{\text{STOP}} \leftarrow 0$ 
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    end if
  else ▷ Final possibility: the unit is OFF.
     $S_{\text{OFF}} \leftarrow 1$ 
     $S_{\text{STOP}} \leftarrow 0$ 
     $S_{\text{START}} \leftarrow 0$ 
     $S_{\text{ON\_UP}} \leftarrow 0$ 
     $S_{\text{ON\_DOWN}} \leftarrow 0$ 
     $S_{\text{ON\_FLAT}} \leftarrow 0$ 
  end if
end for
```

boxes drawn from v2 evidence coords:

red = biggest raster/appearance diff (ink_f1, ssim)

amber = most-displaced matched word (center_q90)

ARM A v0.1 gates 1/4

G1=P G2=F G3=F G4=F ink=0.56 ltsim=0.85 cq90=0.03 numF1=0.989

→ numbers wrong: missing 1

→ text not fully preserved (token recall/precision < .95)

ARM B v2 gates 4/4

G1=P G2=P G3=P G4=P ink=0.49 ltsim=0.81 cq90=0.06 numF1=1.000

→ all gates pass; residual is visual polish (amber/red boxes)

REFERENCE (pdfLaTeX)

ARM A: v0.1 scalar feedback

ARM B: v2 evidence-vector feedback

NO.	name	tR (min)	Exp.	tR MLR	tR ANN	NO.	name
m1	Dichlorodifluoromethane	1.35		1.50	1.52	m44	p-Xylene
m2	Carbon disulfide	4.11		2.71	2.67	m45	m-Xylene
m3	Allyl chloride	4.11		6.57	6.57	m46	o-Xylene
m4	Methylene chloride	4.40		3.93	4.12	m47	Styrene
m5	1,1-Dichloroethene	4.57		6.75	6.79	m48	Bromoforb
m6	Acetone	4.57		3.66	4.22	m49	Isopropylbenzene (Cumene)
m7	trans-1,2-Dichloroethene	4.57		8.77	8.76	m50	cis-1,4-Dichloro-2-butene
m8	Acrylonitrile	5.00		4.52	4.80	m51	Trichlorofluoromethane
m9	1,1-Dichloroethane	6.14		8.62	8.16	m52	1,1,2,2-Tetrachloroethane
m10	Vinyl acetate	6.43		11.47	11.50	m53	Bromobenzene
m11	2,2-Dichloropropane	8.10		12.42	12.61	m54	1,2,3-Trichloropropane
m12	Chloromethane	1.49	-0.34	-0.12		m55	n-Propylbenzene
m13	cis-1,2-Dichloroethene	8.25		9.82	9.86	m56	2-Chlorotoluene
m14	Propionitrile	8.51		5.61	5.99	m57	trans-1,4-Dichloro-2-butene
m15	Chloroform	9.01		7.78	7.50	m58	1,3,5-Trimethylbenzene
m16	Methacrylonitrile	9.19		8.47	8.74	m59	4-Chlorotoluene
m17	1,1,1-Trichloroethane	10.18		13.58	12.60	m60	Pentachloroethane
m18	Carbon tetrachloride	11.02		11.75	11.23	m61	1,2,4-Trimethylbenzene
m19	1,1-Dichloropropene	11.50		14.4	14.29	m62	Acrolein
m20	Benzene	1.56	1.70	1.70	1.84	m63	sec-Butylbenzene
m21	Vinyl Chloride	12.09		11.39	11.53	m64	tert-Butylbenzene
m22	1,2-Dichloroethane	14.03		13.84	13.67	m65	p-Isopropyltoluene
m23	Trichloroethene	14.51		15.14	15.43	m66	1,3-Dichlorobenzene
m24	1,2-Dichloropropane	15.39		16.45	16.45	m67	1,4-Dichlorobenzene
m25	Dibromomethane	15.43		12.66	12.21	m68	Benzyl chloride
m26	methacrylatea Methyl	15.50		15.79	15.69	m69	n-Butylbenzene
m27	1,4-Dioxane	16.17		15.46	15.43	m70	1,2-Dichlorobenzene
m28	2-Chloroethyl vinyl ether	17.32		15.02	15.92	m71	1,2-Dibromo-3-chloropropane
m29	4-Methyl-2-pentanone	17.47		16.13	16.81	m72	1,2,4-Trichlorobenzene
m30	trans-1,3-Dichloropropene	2.19		2.08	1.94	m73	Iodomethane
m31	Bromomethane	18.29		18.63	18.53	m74	Hexachlorobutadiene
m32	Toluene	19.38		15.38	15.90	m75	Naphthalene
m33	cis-1,3-Dichloropropene	19.59		19.34	19.50	m76	1,2,3-Trichlorobenzene
m34	methacrylatea Ethyl	20.01		24.52	24.70	m77	1,4-Difluorobenzene
m35	2-Hexanone	20.30		21.03	21.24	m78	Chlorobenzene-d5
m36	Tetrachloroethene	20.26		19.88	19.62	m79	1,4-Dichlorobenzene-d4
m37	1,3-Dichloropropane	20.51		16.58	16.67	m80	4-Bromofluorobenzene
m38	Dibromochloromethane	21.19		21.94	21.62	m81	1,2-Dichlorobenzene-d4a
m39	1,2-Dibromomethane	21.52		20.89	20.21	m82	Dichloroethane-d4
m40	Chloroethane	2.21		2.83	3.03	m83	Acetonitrile

raster diff							
NO.	name	tR (min)	Exp.	tR MLR	tR ANN	NO.	name
m1	Dichlorodifluoromethane	1.35		1.50	1.52	m44	p-Xylene
m2	Carbon disulfide	4.11		2.71	2.67	m45	m-Xylene
m3	Allyl chloride	4.11		6.57	6.57	m46	o-Xylene
m4	Methylene chloride	4.40		3.93	4.12	m47	Styrene
m5	1,1-Dichloroethene	4.57		6.75	6.79	m48	Bromoforb
m6	Acetone	4.57		3.66	4.22	m49	Isopropylbenzene (C
m7	trans-1,2-Dichloroethene	4.57		8.77	8.76	m50	cis-1,4-Dichloro-2-bu
m8	Acrylonitrile	5.00		4.52	4.80	m51	Trichlorofluorometh
m9	1,1-Dichloroethane	6.14		8.62	8.16	m52	1,1,2,2-Tetrachloroeth
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m20	Benzene	1.56		1.70	1.84	m63	sec-Butylbenzene
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m24	1,2-Dichloropropane	15.39		16.45	16.45	m67	1,4-Dichlorobenzene
m25	Dibromomethane	15.43		12.66	12.21	m68	Benzyl chloride
m26	methacrylates Methyl	15.50		15.79	15.69	m69	n-Butylbenzene
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m28	2-Chloroethyl vinyl ether	17.32		15.02	15.92	m71	1,2-Dibromo-3-chlorop
m29	4-Methyl-2-pentanone	17.47		16.13	16.81	m72	1,2,4-Trichlorobenzene
m30	trans-1,3-Dichloropropene	2.19		2.08	1.94	m73	Iodomethane
m31	Bromomethane	18.29		18.63	18.53	m74	Hexachlorobutadieneb
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m37	1,3-Dichloropropane	20.51		16.58	16.67	m80	4-Bromofluorobenzene
m38	Dibromochloromethane	21.19		21.94	21.62	m81	1,2-Dichlorobenzene-d
m39	1,2-Dibromomethane	21.52		20.89	20.21	m82	Dichloroethane-d4
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m5	1,1-Dichloroethene	4.57	6.75	6.79	m48	Bromoforb	
m6	Acetone	4.57	3.66	4.22	m49	Isopropylbenzene (Cumene)	
m7	trans-1,2-Dichloroethene	4.57	8.77	8.76	m50	cis-1,4-Dichloro-2-butene	
m8	Acrylonitrile	5.00	4.52	4.80	m51	Trichlorofluoromethane	
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boxes drawn from v2 evidence coords:

red = biggest raster/appearance diff (ink_f1, ssim)

amber = most-displaced matched word (center_q90)

ARM A v0.1 gates 3/4

G1=P G2=P G3=P G4=F ink=0.59 ltsim=0.90 cq90=0.02 numF1=0.997

→ numbers wrong; missing 1

ARM B v2 gates 4/4

G1=P G2=P G3=P G4=P ink=0.67 ltsim=0.89 cq90=0.01 numF1=1.000

→ all gates pass; residual is visual polish (amber/red boxes)

REFERENCE (pdfLaTeX)

ARM A: v0.1 scalar feedback

ARM B: v2 evidence-vector feedback

I. INTRODUCTION

In Ref. [1] two different expressions of quantum dual total correlation were obtained: one in terms of von Neumann entropy and other in terms of quantum relative entropy. It was claimed that the two expressions are equivalent. In a comment [2] on Ref. [1], Lee *et al.* have shown that the quantum dual total correlation of an n -partite quantum state cannot be represented as the quantum relative entropy between $(n-1)$ copies of the quantum state and the product of n different reduced quantum states for $n \geq 3$. They arrived at this conclusion by considering explicitly the “support” condition of quantum relative entropy. Essentially, what Lee *et al.* have shown is that the following two expressions are not equal for $n \geq 3$:

$$I_n(\rho) := \sum_{k=1}^n S(\rho_k) - (n-1)S(\rho), \quad (1)$$

where $\rho_k = \text{tr}_k(\rho)$ denotes the $(n-1)$ -partite quantum state obtained by taking the partial trace on the k^{th} party of ρ , and

$$J_n(\rho) := S(\rho^{\otimes(n-1)} \parallel \otimes_{k=1}^n \rho_k), \quad (2)$$

where the quantum relative entropy is

$$S(\tau \parallel \sigma) := \begin{cases} \text{tr}(\tau \log \tau) - \text{tr}(\tau \log \sigma) & \text{if } \text{supp}(\tau) \subseteq \text{supp}(\sigma) \\ \infty & \text{otherwise.} \end{cases}$$

To justify their claim, authors provide two examples which imply that the above two expressions of n -partite quantum mutual information are not equivalent. $I_n(\rho)$ in Eq. (1) is non-negative and non-increasing under local CPTP maps [3], and therefore is a suitable monotonic measure of multi-partite correlations, while $J_n(\rho)$ in Eq. (2) is not.

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The claim of Lee *et al.* is right. In this article we show analytically why the above two expressions are not equivalent. We begin with expression of $I_n(\rho)$ [Eq. (1)] and proceed to show that this is not equal to $J_n(\rho)$ [Eq. (2)], as argued below.

raster diff

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boxes drawn from v2 evidence coords:

red = biggest raster/appearance diff (ink_f1, ssim)

amber = most-displaced matched word (center_q90)

ARM A v0.1 gates 1/4

G1=P G2=F G3=F G4=F ink=0.37 ltsim=0.76 cq90=0.18 numF1=0.992

→ numbers wrong: extra 3,3

→ text not fully preserved (token recall/precision < .95)

ARM B v2 gates 1/4

G1=P G2=F G3=F G4=F ink=0.34 ltsim=0.77 cq90=0.21 numF1=0.992

→ numbers wrong: extra 3,3

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REFERENCE (pdfLaTeX)

ARM A: v0.1 scalar feedback

ARM B: v2 evidence-vector feedback

Equation Sample 1

Dataset-expansion sample

1 Derivation

The following display is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

$$\begin{aligned} \alpha_3 &\stackrel{(i)}{\leq} \sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N \min\{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho), \frac{1}{S(A+B)}\}}} \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-) \\ &\leq \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N}} \sum_{s \in S, a \in A} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sum_{s \in S, a \in A} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \cdot \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)}} \\ &\stackrel{(ii)}{\leq} \sqrt{\frac{C_{\text{clipped}}^*}{N} \log \frac{N}{\delta}} \cdot \sqrt{SA} \cdot \sqrt{\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \left[\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \right] \sqrt{\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\leq 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sqrt{\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\stackrel{(iii)}{=} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B)}{N} \log \frac{N}{\delta}} \sqrt{\sum_{s \in S} d^{\mu^*, \nu_0}(s; \rho) \frac{\mathbb{E}_{a \sim \mu^*(s), b \sim \nu_0(s)}[\text{Var}_{P_{s,a,b}}(V_{\text{pe}}^-)]}{\mathbb{E}_{a \sim \mu^*(s), b \sim \nu_0(s)}[\text{Var}_{P_{s,a,b}}(V_{\text{pe}}^-)]}} \\ &\stackrel{(iv)}{\leq} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B)}{N} \log \frac{N}{\delta}} \sqrt{\sum_{s \in S} d^{\mu^*, \nu_0}(s; \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)}. \end{aligned}$$

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1 Derivation

The following display is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

$$\begin{aligned} \alpha_3 &\stackrel{(i)}{\leq} \sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \sqrt{\frac{\text{moved 'p' } C_{\text{clipped}}^* \log \frac{N}{\delta}}{N \min\{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho), \frac{1}{S(A+B)}\}}} \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-) \\ &\leq \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N}} \sum_{s \in S, a \in A} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sum_{s \in S, a \in A} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \cdot \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)}} \\ &\stackrel{(ii)}{\leq} \sqrt{\frac{C_{\text{clipped}}^*}{N} \log \frac{N}{\delta}} \cdot \sqrt{SA} \cdot \sqrt{\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \left[\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \right] \sqrt{\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\leq 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sqrt{\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\stackrel{(iii)}{=} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B)}{N} \log \frac{N}{\delta}} \sqrt{\sum_{s \in S} d^{\mu^*, \nu_0}(s; \rho) \frac{\mathbb{E}_{a \sim \mu^*(s), b \sim \nu_0(s)}[\text{Var}_{P_{s,a,b}}(V_{\text{pe}}^-)]}{\mathbb{E}_{a \sim \mu^*(s), b \sim \nu_0(s)}[\text{Var}_{P_{s,a,b}}(V_{\text{pe}}^-)]}} \\ &\stackrel{(iv)}{\leq} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B)}{N} \log \frac{N}{\delta}} \sqrt{\sum_{s \in S} d^{\mu^*, \nu_0}(s; \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \text{moved 's' } \end{aligned}$$

Equation Sample 1

Dataset-expansion sample

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The following display is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

$$\begin{aligned} \alpha_3 &\stackrel{(i)}{\leq} \sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \sqrt{\frac{\text{moved 'p' } C_{\text{clipped}}^* \log \frac{N}{\delta}}{N \min\{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho), \frac{1}{S(A+B)}\}}} \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-) \\ &\leq \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N}} \sum_{s \in S, a \in A} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sum_{s \in S, a \in A} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \cdot \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)}} \\ &\stackrel{(ii)}{\leq} \sqrt{\frac{C_{\text{clipped}}^*}{N} \log \frac{N}{\delta}} \cdot \sqrt{SA} \cdot \sqrt{\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \left[\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \right] \sqrt{\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\leq 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sqrt{\sum_{s \in S, a \in A} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\ &\stackrel{(iii)}{=} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B)}{N} \log \frac{N}{\delta}} \sqrt{\sum_{s \in S} d^{\mu^*, \nu_0}(s; \rho) \frac{\mathbb{E}_{a \sim \mu^*(s), b \sim \nu_0(s)}[\text{Var}_{P_{s,a,b}}(V_{\text{pe}}^-)]}{\mathbb{E}_{a \sim \mu^*(s), b \sim \nu_0(s)}[\text{Var}_{P_{s,a,b}}(V_{\text{pe}}^-)]}} \\ &\stackrel{(iv)}{\leq} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B)}{N} \log \frac{N}{\delta}} \sqrt{\sum_{s \in S} d^{\mu^*, \nu_0}(s; \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \text{moved 's' } \end{aligned}$$

boxes drawn from v2 evidence coords:

red = biggest raster/appearance diff (ink_f1, ssim)

amber = most-displaced matched word (center_q90)

ARM A v0.1 gates 1/4

G1=P G2=F G3=F G4=F ink=0.44 ltsim=0.74 cq90=0.08 numF1=0.923

→ numbers wrong: missing 1

→ text not fully preserved (token recall/precision < .95)

ARM B v2 gates 2/4

G1=P G2=F G3=F G4=P ink=0.32 ltsim=0.74 cq90=0.05 numF1=1.000

→ text not fully preserved (token recall/precision < .95)

1. Non-compensatory feedback helps, consistently: across 6 hard samples Arm B (v2) matched or beat Arm A on the HARD GATES a human objects to, winning outright in 5/6 (avg 3.2/4 vs 2.0/4). It closed exact-number-F1 to 1.000 in 5/6, where Arm A stalled at 0.989-0.998 — the gate ladder names the miss, so the agent fixes it.
2. Arm A (v0.1 scalar) tends to buy visual polish (edges Arm B on the drivers in 4/6, by ~ 0.02 -0.08) while letting a single wrong number or off-by-one page slip inside the weighted average — the compensatory failure v2 warned of.
3. Content vs visual are near-independent axes (Phase A: $r(\text{strict_f1}, \text{overall}) = +0.20$); one blended number cannot represent both, which is exactly why the v2 vector is more actionable feedback.
4. Strict token gates are HARDER, not impossible, on math/algorithm PDFs: on 09_algorithms_003, gate-first feedback pushed Arm B from 1/4 to 4/4 (token recall/precision cleared .95) where Arm A stayed at 1/4. Kept strict per request.
5. The verbose full paper (arxiv5t_paper_019) is the shared ceiling (both 1/4) — real multi-page content divergence neither feedback resolves at \$2 / medium effort.

Caveat: $n=6$, one seed per cell (06_tables repeated across 2 seeds agreed: A 3/4, B 4/4). Mechanism evidence, not a leaderboard; no human ratings exist. All arms graded by pdf_metric_axes_v2 @96 DPI; v2/gates/driver-reward untouched.